SHEN GE

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PROFESSIONAL SUMMARY

Senior ML/AI Engineer with extensive experience in designing and deploying production ML and LLM systems for fraud detection, anti-money laundering, and financial risk. Expertise in multi-agent LLM orchestration, Machine Learning and Deep Learning, Graph ML (GNN, community detection), and end-to-end model lifecycle management.

PROFESSIONAL EXPERIENCE

PayPal — Senior AI Engineer

Shanghai, China (Feb 2019 – Aug 2025) | Singapore (Sep 2025 – Present)

AML Investigation Mate — Multi-Agent LLM System *(Sep 2025 – Present)*

- Architected a LangChain deep-agent multi-agent system (Main agent with plan-and-execute + ReAct-style specialized sub-agents) automating end-to-end AML case investigation — evidence retrieval, regulatory rule reasoning, risk screening, and report generation — achieving 80% decision accuracy, on par with senior investigators.
- Built a Graph-RAG knowledge layer encoding 50+ internal SOPs and external AML regulations into a structured knowledge graph (Condition → Action → Legal Document), enabling grounded agent reasoning.
- Reduced average case review time from 3–4 hours to under 30 minutes (85% reduction) across 15 investigators via DSPy automatic prompt optimization and iterative multi-agent evaluation pipelines.
- Enabled investigators to interactively query the agent for case details, drill into specific evidence, modify analysis parameters, and iteratively refine report drafts through a multi-turn conversational interface.

Text-to-SQL Investigation Toolkit for Buyer Risk Operation — LLM-Powered Query Generation *(Jan 2025 – Aug 2025)*

- Designed a multi-step RAG pipeline (query rephrase → BGE-M3 embedding retrieval → Chain-of-Thought SQL generation) converting natural language to BigQuery SQL with 94% correctness, reducing manual SQL coding time by 80%.
- Built a multi-agent workflow via LangGraph, integrating intent prediction, user-input clarification, error self-healing, interactive SQL refinement, data visualization, and feedback reuse into a unified human-in-the-loop investigation platform.

Proactive Fraud Detection — Graph-Based Account Linking & Clustering *(Jun 2023 – Dec 2024)*

- Led end-to-end development of a graph-based fraud detection pipeline over 100M+ accounts and 300M edges, evolving through 4 clustering generations (seed-based → fuzzy asset combination → Louvain community detection → embedding similarity) to surface fraud rings early in the account lifecycle — delivering $20M+ in annual net-loss savings.
- Trained an AutoEncoder via model distillation to extract account embeddings, then applied edge-aware GAT (DGL/PyTorch) — concatenating edge features (linking type, asset riskiness) to attention outputs for full signal preservation — computing group-level and account-level risk scores to empower manual review and lead-automation strategies. Formulated investigator resource allocation as Integer Linear Programming (ILP) to increase review efficiency.

Machine Learning Model Development — Multi-Domain Risk Modeling *(Feb 2019 – May 2023)*

- Owned end-to-end model lifecycle across multiple domains (stolen financial instruments, account takeover, buyer AUP violation, collusion, merchant website compliance, dispute automation), adapting modeling strategies to each domain's unique constraints — from tagging design and feature engineering through training, evaluation, and production deployment.
- Applied a wide range of methodologies tailored to business needs: PU-Learning for incomplete-label expansion, custom loss functions and sample weighting to prioritize high-risk scores in DNNs, LSTM encoder-decoder for user behavior sequence anomaly detection, Word2Vec + KMeans for buyer behavioral clustering, BERT embeddings for transaction note semantic features, and LightGBM/XGBoost as rapid baselines with iterative NN improvements.
- Beyond traditional modeling, developed unstructured data acquisition, domain feature engineering, and explainable compliance delivery including: Built a merchant-website risk-scoring model from unstructured HTML and external traffic data; designed custom feature engineering pipelines (asset riskiness, device fingerprint, IP geo-anomaly, abnormal email patterns, etc.); delivered explainable AI (SHAP) outputs for stakeholder trust and regulatory compliance across all domains.

Leadership: Mentored 2 junior data scientists on ML development and fraud domain expertise. Led PayPal's Recommendation System Study Group (2024), driving cross-team knowledge sharing.

TECHNICAL SKILLS

LLM & Agents: LangChain, LangGraph, DSPy, Multi-Agent system
ML & Deep Learning: PyTorch, DGL, LightGBM, XGBoost, SHAP, Pandas, Scikit-learn
Data & Infrastructure: Python, SQL, BigQuery, Spark, Airflow, Docker, GCP

EDUCATION

The Chinese University of Hong Kong — MSc in Business Analytics | 2017 – 2018 | GPA: 3.40/4.0

College of William & Mary — B.S. in Applied Mathematics (CS Minor) | 2013 – 2017 | GPA: 3.62/4.0, 6× Dean's List

葛 燊

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个人简介

高级 ML/AI 工程师，拥有机器学习及大语言模型系统的设计与落地经验，业务覆盖欺诈检测、反洗钱及金融风控领域。掌握多智能体 LLM 系统编排、机器学习与神经网络、图神经网络（GNN、社区发现）以及模型全生命周期管理。

工作经历

PayPal — 高级 AI 工程师

上海 (2019.02 – 2025.08) | 新加坡 (2025.09 – 至今)

AML 案件智能调查系统 — 多智能体 LLM 系统 *(2025.09 – 至今)*

- 基于 LangChain Deep Agent 设计并搭建多智能体架构（主智能体采用“规划-执行”（plan-and-execute）模式，搭配 ReAct 专业化子智能体），实现反洗钱案件的端到端自动化调查流程——涵盖证据检索、法规规则推理、风险筛查以及调查报告生成——决策准确率达 80%，与资深调查员的判断水平持平。
- 构建 Graph-RAG 知识层，将超过 50 份内部 SOP 以及外部反洗钱法律法规编码为结构化知识图谱（触发条件 → 处理操作 → 法律条文），使智能体具备基于合规要求的推理能力。
- 借助 DSPy 库的 prompt 优化算法实现提示词的自动化优化，并搭建迭代评估流程持续改进智能体表现，最终将单案审查时间从 3–4 小时缩短至 30 分钟以内（效率提升约 85%），该结果经 15 名调查员实际使用验证。
- 支持调查员与智能体进行全流程多轮交互式对话，可深入查询案件细节、调整分析参数、逐步修改和完善调查报告草稿，整个过程保留完整的反馈轨迹并生成轨迹报告，驱动后续持续优化。

自然语言转 SQL 案件调查工具 (Buyer Risk) — 基于 LLM 的 Text-to-SQL 生成 *(2025.01 – 2025.08)*

- 设计多步骤检索增强生成流水线（先对用户查询进行意图识别和语义改写，再通过 BGE-M3 向量检索相似查询样例，最后结合思维链提示生成 BigQuery SQL），生成准确率达 94%，使调查员人工编写 SQL 的时间减少 80%。
- 基于 LangGraph 构建多智能体工作流，将意图识别、输入改写、错误自动修复反馈循环、交互式 SQL 修改、PandasAI 数据可视化分析以及历史反馈复用等功能整合为一个统一的人机协同调查平台。

主动欺诈检测 — 基于图的账户关联与聚类 *(2023.06 – 2024.12)*

- 主导开发基于图的欺诈检测解决方案，日均处理规模达 1 亿以上账户、3 亿条关联边，方案先后经历 4 代聚类方法的演进（种子账户图遍历 → 模糊关联特征组合 → Louvain 社区发现算法 → 账户嵌入向量相似度计算），在账户生命周期早期主动识别欺诈团伙——年化净损失节省超过 $20M。
- 通过模型蒸馏技术训练 AutoEncoder 提取账户级别的嵌入向量表示，然后在关联图上应用边感知 GAT（DGL/PyTorch）——将边特征（关联类型、资产风险度）直接拼接至注意力层输出以完整保留边信息——同时计算组级和账户级风险评分，为人工审核和自动化决策策略提供依据。此外，将调查员资源分配问题建模为整数线性规划（ILP）优化求解，提升审查效率。

风控领域 ML 模型开发 — 多场景风险建模 *(2019.02 – 2023.05)*

- 端到端负责多个风控业务场景的模型全生命周期管理（盗刷、盗号、买家违规行为、串谋欺诈、商户网站合规审查、交易争议自动化等），根据每个领域的业务特点和数据特征灵活选择和调整建模策略——从标签体系设计、特征工程到模型训练、评估与生产部署。
- 根据不同业务需求灵活运用多种建模方法：如通过 PU-Learning 解决正样本标签不完整的问题；设计神经网络的损失函数加权与样本权重平衡策略，使模型优先学习高损失欺诈样本；使用 LSTM 编解码器对用户行为序列进行异常检测；借助 Word2Vec 生成买家行为嵌入并结合 KMeans 进行行为聚类；使用 BERT 嵌入抽取交易备注的语义特征；以 LightGBM/XGBoost 作为快速基线模型并通过神经网络迭代优化。
- 除传统建模外，还负责非结构化数据采集、领域特征工程与可解释合规交付的端到端搭建。其中包括：构建商户网站风控评分系统，基于爬取的非结构化 HTML 页面内容和外部流量数据进行风险评估；设计定制化特征工程流水线（Asset 风险评分、设备指纹识别、IP 地理位置异常检测、可疑注册邮箱识别等）；交付基于 SHAP 的可解释 AI 输出，满足内部审查和合规要求。

团队管理： 指导 2 名初级数据科学家，帮助其快速掌握 ML 模型开发流程和风控业务知识。组织并主导 PayPal 推荐系统学习小组（2024），推动跨团队的技术分享与知识交流。

技术技能

LLM 与智能体： LangChain, LangGraph, DSPy

机器学习与深度学习： PyTorch, DGL, LightGBM, XGBoost, SHAP, Pandas, Scikit-learn

数据与基础设施： Python, SQL, BigQuery, Spark, Airflow, Docker, GCP

教育背景

香港中文大学 — 商业分析硕士 | 2017 – 2018 | GPA: 3.40/4.0

威廉与玛丽学院 (College of William & Mary) — 应用数学学士（计算机辅修） | 2013 – 2017 | GPA: 3.62/4.0，6× Dean's List